

		$= 25000$
22	A	Algebraic sum of potential due to charges at A and B is zero, so $\frac{1}{4\pi\epsilon_0} \frac{+2.4 \times 10^{-6}}{7.0 \text{ cm}} + \frac{1}{4\pi\epsilon_0} \frac{-2.9 \times 10^{-6}}{\text{BX cm}} = 0$ giving BX = 8.458 cm Pythagoras theorem gives $L = \sqrt{8.458^2 - 7.0^2} = 4.74 \text{ cm}$
22	D	In the circuit, $V_{DO} = 2 / (2 + 4 + 5 + 22) \times 22.0 = 1.24 \text{ V}$